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NATIONAL INSTITUTE OF TECHNOLOGY PATNA
END-SEMESTER EXAMINATION, JAN-JUNE 2025

Program: B.Tech/B.Tech-DD
Department: CSE
Full Marks: 60
Duration of Examination: 3 Hours

Semester: 4th
Course Code: CS44113
Course Name: Computer Networks

INSTRUCTIONS

- *Please mention your section at the top of the answer script.*
- *Answer all the questions. Assume Missing data, if any.*
- *Each question includes the corresponding marks, Course Outcome (CO), and Bloom's Level (BL).*

1. A. Analyze the working of the Dynamic Host Configuration Protocol (DHCP) in a typical network environment. Explain each stage of the DHCP process in detail. Discuss what happens if multiple DHCP servers respond to a single client's request. *(8M) [CO6][BL4]*

   B. In the CRC (Cyclic Redundancy Check) method, the dataword to be transmitted is 1101011011. The generator polynomial is given as $G(x) = x^4 + x + 1$, which corresponds to the binary divisor 10011. After applying the CRC encoding process, what is the final encoded frame that will be transmitted from the sender? (Show all steps of binary division clearly) *(4M) [CO3][BL3]*

2. A. Explain the 3-way handshaking for connection establishment in TCP connection with a neat diagram. *(6M) [CO5][BL2]*

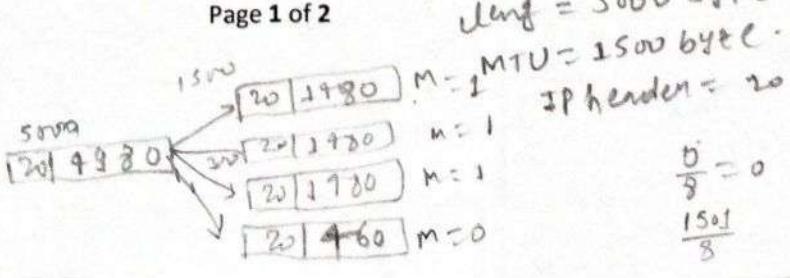
   B. In a reliable data transfer system with a sender window size of 4, the sender transmits packets 0, 1, 2, and 3, but all the acknowledgment packets from the receiver are lost. *(2+2+1+1 M) [CO5][BL3]*
   i. Describe how the sender behaves in this scenario under the Go-Back-N (GBN) protocol.
   ii. Describe how the sender behaves under the Selective Repeat (SR) protocol.
   iii. In each case, explain how the receiver handles the retransmitted packets.
   iv. Does any form of deadlock occur in either protocol under these conditions? Justify your answer.

3. A. A data packet of size 5000 bytes is to be sent over a network with an MTU (Maximum Transmission Unit) of 1500 bytes. The IP header size is 20 bytes. *(6M)[CO4][BL3]*
   i. How many fragments will be generated, and what will be the size of each fragment, including the IP header? Assume that the packet is fragmented at the network layer.
   ii. Calculate the fragment offset and the flags (More Fragments flag) for each fragment.

   B. You're assigned the network: 172.16.0.0/24. You need to subnet this network to meet the following requirements:

Department	Hosts Required
R&D	60
IT	40
Sales	30
Legal	25
HR	15

Use VLSM to assign subnet blocks to each department from the 172.16.0.0/24 block. For each department, list Subnet Address, CIDR Notation, Usable Host IP Range, Broadcast Address, mention how many total IP addresses are left unused after subnetting. *(6M)[CO4][BL3 BL4]*



4. A. Analyze the process of framing at the Data Link Layer and explain its importance in reliable data transmission. Describe the different framing methods briefly, highlighting how each method identifies frame boundaries. Finally, evaluate which of these methods is most suitable for communication over noisy channels, justifying your choice. *(6M)[CO3] [BL4]*

B. A Slotted ALOHA network uses a shared channel with a bandwidth of 1 Mbps to transmit 1500-bit frames. Calculate the throughput for the system (all stations together) under the following conditions:

i. When the system produces 1200 frame attempts per second.
ii. When the system produces 800 frame attempts per second.
iii. When the system produces 400 frame attempts per second. *(6M)[CO3] [BL2 BL3]*

5. A. Apply your understanding of transmission modes to explain how simplex, half-duplex, and full-duplex communication would function in the following scenarios: (i) a TV broadcast, (ii) a walkie-talkie communication system, and (iii) a telephone call. Justify your answers with proper reasoning. *(6M) [CO2][BL4, BL5]*

B. Explain how the Routing Information Protocol (RIP) works, including its basic operation and how it calculates the shortest path. What are the key limitations of RIP? *(6M) [CO4][BL2]*

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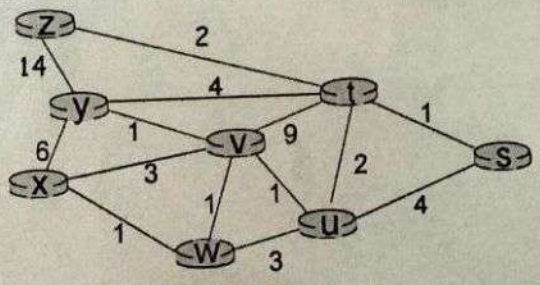
B Tech: Semester-IV

Course Name: Computer Networks **Course Code: CS44113**

Maximum Time: 3 hours **Max. Marks: 60**

Instructions:

1. Attempt all questions.
2. Assume any suitable data, if necessary. *(Any other instruction need to provide by the concern faculty)*
3. The marks, Course Outcome (CO) and Bloom's Level (BL) related to questions are mentioned on the right-hand side margin.

		Marks	CO	BL
1	Develop an algorithm for CSMA/CD method with suitable flow diagram/example? How it can be differ from Pure ALOHA protocol. One hundred stations on a slotted ALOHA network share a 1-Mbps channel. If frames are 1000 bits long, determine the throughput if each station is sending 10 frames per second.	3+3+4 = 10	CO-3	Applying and Evaluating
2	List the difference (at least five) between IPv4 and IPv6 protocol with suitable diagram? An ISP is granted a block of addresses starting with 120.60.4.0/20. The ISP wants to distribute these blocks to 100 organizations with each organization receiving 8 addresses only. Design the sub blocks and give the slash notation for each sub-block. Find out how many addresses are still available after these allocations.	5+5 = 10	CO-4	Remembering and Creating
3	Discuss the algorithmic steps for link state routing algorithm with suitable diagram/example? How does it differ (at least five) from distance vector routing algorithm? Consider the network shown below (Fig-1) and assume that each node initially knows the costs to each of its neighbors. Consider the link state routing algorithm and create the distance table entries at node X (Source node).  Figure 1: The Network	3+2+5 = 10	CO-4	Applying
4	Compare and contrast TCP segment with UDP datagram in detail along with flow and congestion control mechanism for TCP? The following is a dump of a TCP header in hexadecimal format: $(05320017\ 00000001\ 00000000\ 500207FF\ 00000000)_{16}$. Determine the following: a) Destination port number, b) Sequence number, c) Length of the header, d) Type of the segment, and e) Window size	5+5 = 10	CO-5	Understanding and Evaluating
5	Write an effective note on any four with suitable diagram (s) a) Traffic shaping algorithms b) Hypertext transfer protocol c) Error detection techniques d) Stop & wait ARQ protocol e) Time division multiplexing f) OSI reference model	20 (5 marks each)	CO-2-CO-6	Creating